

1. Orkhan Khalilzade

$(x + 7)$ is a factor of the quadratic function $y = f(x)$, and $x = c$ is a positive zero of the function. If the minimum of the function is $(\frac{63}{8}, k)$, what is the value of c ?

2. Orkhan Khalilzade

For two acute angles, $\angle A$ and $\angle B$, $\cos A = \sin B$. The measures, in degrees, of $\angle A$ and $\angle B$ are $13x + 10$ and $48 + 3x$, respectively. What is the value of x ?

3. Orkhan Khalilzade

A baseball coach charges each athlete \$150 for the first month and \$95 for each additional month. Which of the following functions correctly represents the total revenue $R(m)$, in dollars, for m months and n athletes, where m and n are positive integers?

- A) $R(m) = 95mn + 55$
- B) $R(m) = 95mn + 150$
- C) $R(m) = 95mn + 55n$
- D) $R(m) = 95mn + 150n$

4. Orkhan Khalilzade

$$y = -3x^2 - 58$$
$$y = px - 46$$

In the given system of equations, p is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) in the xy -plane. Which of the following could be the value of x ?

- A) -2
- B) -4
- C) 12
- D) 16

5. Orkhan Khalilzade

The positive number a is 480% of the number b , and a is 70% of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

- A) 267
- B) 336
- C) 550
- D) 685

6. Orkhan Khalilzade

$$f(x) = a^x - b$$

The exponential function given above passes through points $(c, 8)$ and $(2c, 350)$. What is a possible value of b ?

7. Orkhan Khalilzade

$$f(x) = -7x + b$$

For the given function $f(x)$, which of the following gives the equation for the function $f(x) + 20$, given that the point $(-4, 0)$ lies on the graph of $f(x) - 20$?

- A) $f(x) = -7x - 28$
- B) $f(x) = -7x - 8$
- C) $f(x) = -7x + 8$
- D) $f(x) = -7x + 12$

8. Orkhan Khalilzade

Cylinder	Volume	Surface Area
A	$1,296\pi$	$k\pi$
B	$162,000\pi$	$j\pi$

The table gives the volume of two similar cylinders. If the radius of cylinder A is 9 cm. What is the value of $j - k$?

$$\text{Volume} = \pi r^2 h$$

$$\text{Surface area} = 2\pi r^2 + 2\pi r h$$

9. Orkhan Khalilzade

For the function f , for every increase of $\frac{1}{2}$ in the value of x , the value of $f(x)$ increases by a factor of c , where c is a constant. Which of the following forms of function f displays the value of c as the base or coefficient?

A) $f(x) = 56(2)^{6x}$

B) $f(x) = 56(8)^{2x}$

C) $f(x) = 56(64)^x$

D) $f(x) = 56(4,096)^{\frac{x}{2}}$

10. Orkhan Khalilzade

The quadratic function $ax^2 + 200x + c$ has at least one real solution. What is the greatest possible value of ac ?

11. Orkhan Khalilzade

A polygon has exactly 97 sides. If the measure of each of the 97 interior angles of this polygon is $(180p)^\circ$, what is the value of p ?

12. Orkhan Khalilzade

A certain town has an area of 5.48 square miles. Which of the following is closest to the area, in square yards, of this town? (1 mile = 1,760 yards)

A) 552

B) 964

C) 9,645

D) 16,974,848

13. Orkhan Khalilzade

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

$$7x = 105y - 133$$

A) $x = 15y$

B) $\frac{1}{7}x = 15y$

C) $x = 15y - 19$

D) $\frac{1}{7}x = 15y - 19$

14. Orkhan Khalilzade

In triangle ABC , the sum of the measures of angle A and angle B is 90 degrees. The value of $\sin(A)$ is $\frac{2\sqrt{7}}{8}$. What is the value of $\cos(B)$?

A) $\frac{2\sqrt{7}}{8}$

B) $\frac{2\sqrt{7}}{6}$

C) $\frac{6\sqrt{2}}{14}$

D) $\frac{8}{23\sqrt{2}}$

15. Orkhan Khalilzade

What is the value of $\frac{k}{25}$ if the given equation only has one solution?

$$25|x - 4| = k$$

- A) -5 only
- B) 5 only
- C) 5 or -5
- D) 0

16. Orkhan Khalilzade

In the given system of equations, k is a constant. If the system has no solution, what is the value of k ?

$$\begin{aligned}\frac{3}{8}x + \frac{7}{5}y &= \frac{9}{7} - \frac{7}{5}y \\ \frac{5}{4}x + \frac{9}{4} &= ky + \frac{4}{3}\end{aligned}$$

17. Orkhan Khalilzade

In the xy -plane, the graph of the given equation is a circle. Which point lies on this circle?

$$(x + 7)^2 + (y - 11)^2 = 25$$

- A) $(-7, 11)$
- B) $(11, -7)$
- C) $(\sqrt{11} + 7, \sqrt{14} - 11)$
- D) $(\sqrt{11} - 7, \sqrt{14} + 11)$

18. Orkhan Khalilzade

The area of a triangle is equal to x^2 square inches. The base of the triangle is $5 + 2x$ inches, and the height of the triangle is $x - 2$ inches. What is the value of x ?

- A) 2.5
- B) 2.7
- C) 5
- D) 10

19. Orkhan Khalilzade

A circle in the xy -plane has its center at $(3, -7)$ and has a radius of 12. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

20. Orkhan Khalilzade

Sphere A has a radius of $7x$ and sphere B has a radius of $91x$. The volume of sphere B is how many times the volume of sphere A ?

21. Orkhan Khalilzade

One gallon of resin costs \$79 and will cover 48 square feet of a countertop. The countertops in a given room have a total surface area of k square feet. Which equation represents the cost c , in dollars, of resin needed to cover the countertops twice?

- A) $c = \frac{48k}{79}$
- B) $c = \frac{96k}{79}$
- C) $c = 79\left(\frac{k}{24}\right)$
- D) $c = 79\left(\frac{k}{96}\right)$

22. Orkhan Khalilzade

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

$$\begin{aligned} 10x + 8y &= 15 \\ 25x + 20y &= 37.5 \end{aligned}$$

- A) $(r, -\frac{4r}{5} + \frac{15}{8})$
 B) $(r, \frac{5r}{4} + \frac{4}{5})$
 C) $(-\frac{4r}{5} + \frac{3}{2}, r)$
 D) $(\frac{2r}{5} + 10, -\frac{2r}{5} + 25)$

23. Orkhan Khalilzade

A researcher selected 2 random samples of employees in a large company to estimate the percentage of employees that planned to vote in favor of a new division. Based on the first sample, the researcher estimated that 68% of the employees would vote in favor, with an associated margin of error of 9.7%. Based on the second sample, the researcher estimated that 84% of the employees would vote in favor, with an associated margin of error of 6.7%. Assuming the margins of error were calculated in the same way, which of the following best explains why the second sample obtained a smaller margin of error than the first sample?

- A) The first sample contained fewer employees than the second sample.
 B) The first sample contained more employees than the second sample.
 C) The first sample contained a lower percentage of residents that planned to vote in favor of a new division.
 D) The first sample contained a higher percentage of residents that planned to vote in favor of a new division.

24. Orkhan Khalilzade

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(3, 0)$ and a y -intercept at $(0, -342)$. What is the value of $a + b$?

25. Orkhan Khalilzade

684 is $p\%$ greater than 9. What is the value of p ?

26. Orkhan Khalilzade

Two numbers a and b , are each greater than zero, and the fifth root of a is equal to the cube root of b . For what value of x is a^{8x-4} equal to b ?

27. Orkhan Khalilzade

$$\begin{aligned} (j - k) - 16(m + l) &= 689 \\ (j - k) + 12(m + l) &= 1235 \end{aligned}$$

For the given system of equations, what is the value of $7(j - k)$?